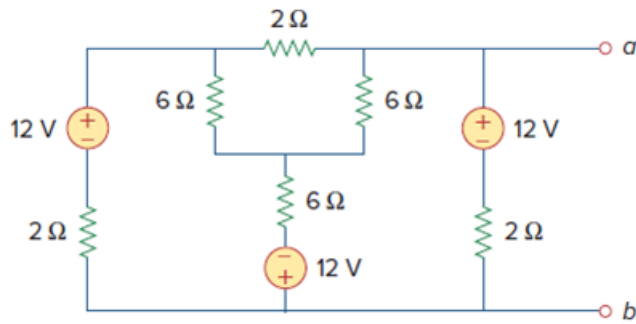


Problem

Obtain the Thevenin and Norton equivalent circuits at terminals $a-b$ of the circuit in Fig. 4.127.

Figure 4.127:



Step-by-step solution

Step 1 of 11

Refer to the Figure 4.127 in the text book.

Draw the following circuit diagram.

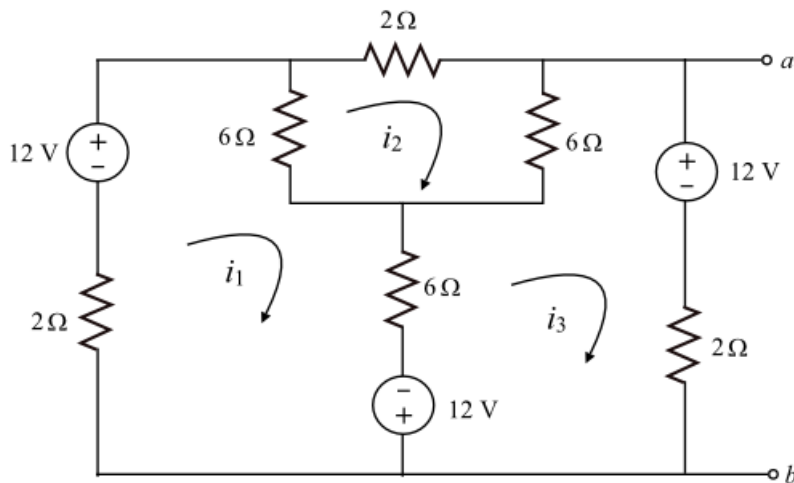


Figure 1

Step 2 of 11

Use mesh analysis, apply Kirchhoff's voltage law in first loop.

$$2i_1 - 12 + 6(i_1 - i_2) + 6(i_1 - i_3) - 12 = 0$$

$$14i_1 - 6i_2 - 6i_3 = 24 \dots\dots (1)$$

Apply Kirchhoff's voltage law in second loop.

$$6(i_2 - i_1) + 2i_2 + 6(i_2 - i_3) = 0$$

$$-6i_1 + 14i_2 - 6i_3 = 0 \dots\dots (2)$$

Apply Kirchhoff's voltage law in third loop.

$$6(i_3 - i_2) + 12 + 2i_3 + 12 + 6(i_3 - i_1) = 0$$

$$-6i_1 - 6i_2 + 14i_3 = -24 \dots\dots (3)$$

Write the equations (1), (2) and (3) in matrix form.

$$\begin{bmatrix} 14 & -6 & -6 \\ -6 & 14 & -6 \\ -6 & -6 & 14 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 24 \\ 0 \\ -24 \end{bmatrix}$$

Step 3 of 11

Use crammers rule to determine the value of current i_3 .

$$\begin{aligned} \Delta &= \begin{vmatrix} 14 & -6 & -6 \\ -6 & 14 & -6 \\ -6 & -6 & 14 \end{vmatrix} \\ &= 14(196 + 36) + 6(84 + 36) - 6(36 + 84) \\ &= 800 \end{aligned}$$

$$\begin{aligned} \Delta_3 &= \begin{vmatrix} 14 & -6 & 24 \\ -6 & 14 & 0 \\ -6 & -6 & -24 \end{vmatrix} \\ &= 14(-336 + 0) + 6(144 + 0) + 24(36 + 84) \\ &= -960 \end{aligned}$$

Determine the value of current i_3 .

$$\begin{aligned} i_3 &= \frac{\Delta_3}{\Delta} \\ &= \frac{-960}{800} \\ &= -1.2 \text{ A} \end{aligned}$$

Step 4 of 11

Determine the value of Thevenin resistance.

$$\begin{aligned} V_{TH} &= 12 + 2i_3 \\ &= 12 - 2(1.2) \\ &= 9.6 \text{ V} \end{aligned}$$

Therefore, the value of Thevenin voltage 9.6 V.

Step 5 of 11

Draw the following circuit diagram to determine the Thevenin resistance.

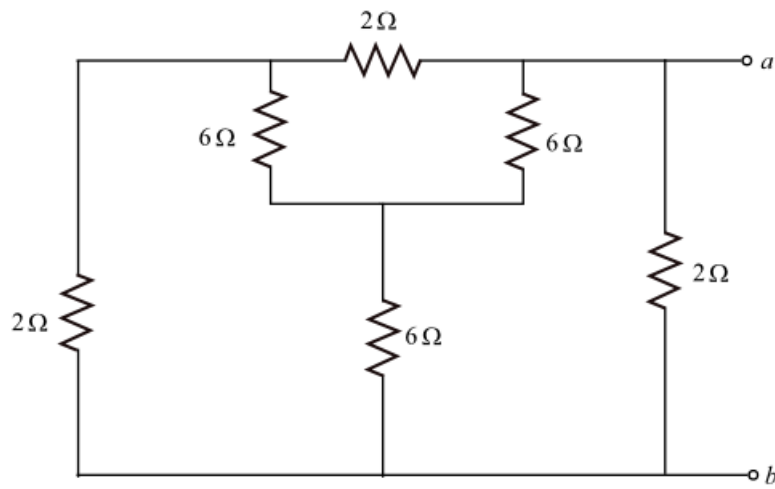


Figure 2

Step 6 of 11

Convert the star connection into delta connection.

$$R_1 = \frac{(6)(6) + (6)(6) + (6)(6)}{6}$$

$$= 18\Omega$$

$$R_2 = \frac{(6)(6) + (6)(6) + (6)(6)}{6}$$

$$= 18\Omega$$

$$R_3 = \frac{(6)(6) + (6)(6) + (6)(6)}{6}$$

$$= 18\Omega$$

Step 7 of 11

Draw the following modified circuit diagram.

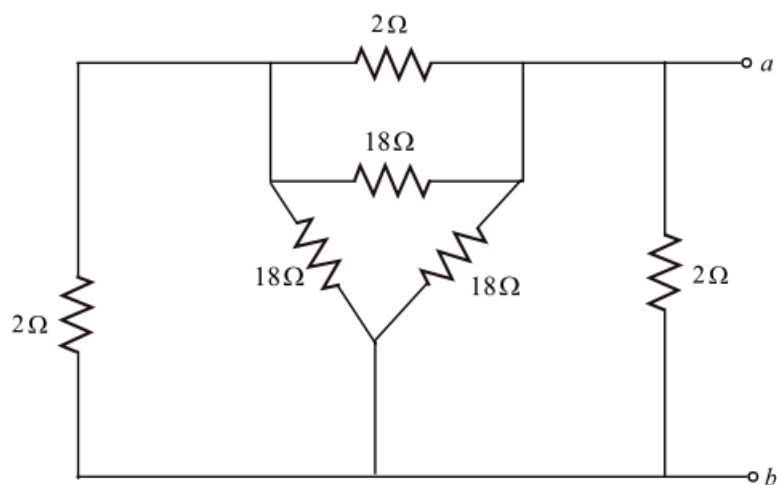


Figure 3

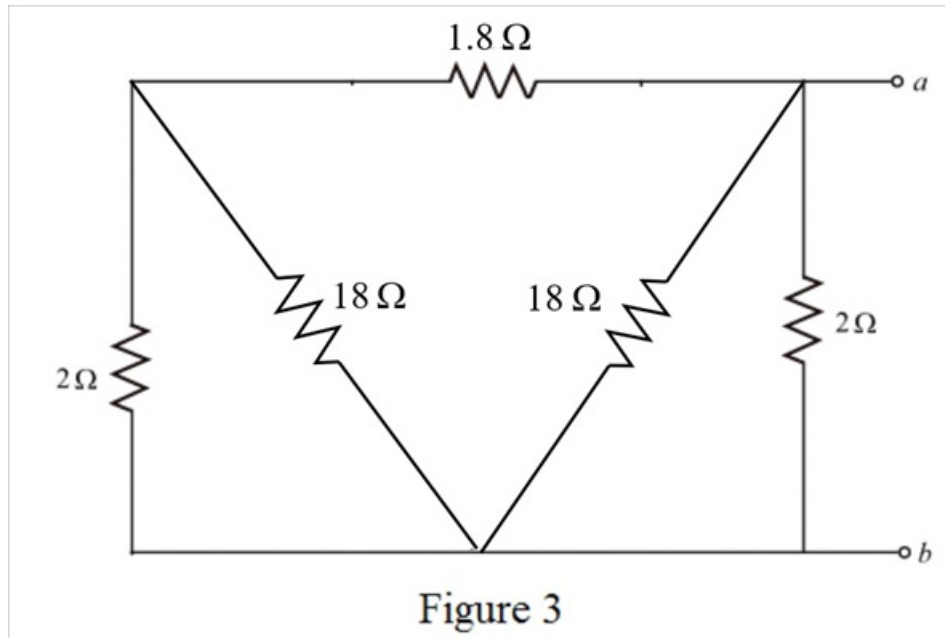
Step 8 of 11

The resistance $18\ \Omega$ and $2\ \Omega$ are in parallel.

$$\begin{aligned} R' &= \frac{18 \times 2}{18 + 2} \\ &= \frac{36}{20} \\ &= 1.8\ \Omega \end{aligned}$$

Step 9 of 11

Draw the simplified circuit.



Step 10 of 11

Determine the equivalent resistance between terminals a and b .

$$\begin{aligned} R_{ab} &= [(18 \parallel 2) + 1.8] \parallel (18 \parallel 2) \\ &= \left[\left(\frac{18 \times 2}{18 + 2} \right) + 1.8 \right] \parallel \left(\frac{18 \times 2}{18 + 2} \right) \\ &= [(1.8) + (1.8)] \parallel (1.8) \\ &= 3.6 \parallel 1.8 \\ &= \frac{3.6 \times 1.8}{3.6 + 1.8} \\ &= 1.2\ \Omega \end{aligned}$$

Therefore, the value of Thevenin resistance is $1.2\ \Omega$.

Step 11 of 11

Determine the Norton's current.

$$\begin{aligned} I_N &= \frac{V_{TH}}{R_{TH}} \\ &= \frac{9.6}{1.2} \\ &= 8 \text{ A} \end{aligned}$$

Therefore, the value of the Norton's current is 8A.